



**Politecnico
di Torino**

Politecnico di Torino

Project 1

Wheel Arch and Ground Clearance Sizing

Car Body Design

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1 Introduction

1.1 Description of the project

The purpose of this report is the design and sizing of the front and rear wheel arches (starting from a top view analysis) and the evaluation of the ground clearance sizing on loading conditions (starting from a side view analysis) of a passenger car under minimum and maximum loading conditions.

The different stages of this report consist of the selection of a commercial vehicle, extracting the relevant data and parameters of this latter. Then the goal is size the front and rear wheel arch, finding the maximum steering angles and design in top and side view. On the last stage it is necessary to estimate the ground clearance changes in maximum and minimum loading conditions.

In this project, the selected vehicle is BMW 3 Series Sedan (2011).



Figure 1: BMW 3 Series Sedan 2011

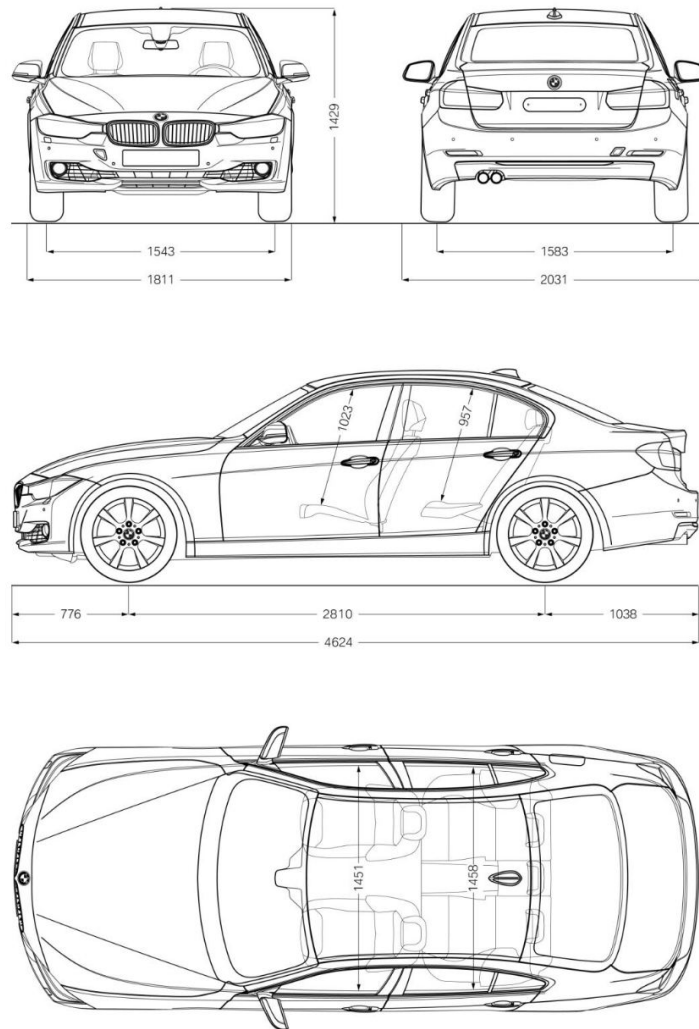


Figure 2: Internal and external dimension

2 Method: detailed description of the desing proce- dures

The relevant data and parameters of the BMW 3 Series (2011) have been chosen from catalogs. The paramaters of the selected vehicle are expressed below:

Parameter	Symbol / Unit	Value
Wheelbase	wb [mm]	2810
Front track	t_f [mm]	1543
Rear track	t_r [mm]	1583
Size of tires	–	255/30 R19
Minimum steering radius	R_1 [mm]	5650
Vehicle mass (empty) (DIN)	m [kg]	1420
Number of occupants	–	5
Weight distribution (F/R)	–	51 / 49

Table 1: Main vehicle parameters – BMW 3 Series Sedan (2011)

2.1 Sizing of the front wheel arch

2.1.1 Maximum steering angles

From a top view analysis, it is possible to find the kinematic steering dimensions (the inner steering angle δ_2 , the outer steering angle δ_1 and steering radius R) with the following equations:

$$\begin{cases} \delta_1 = \tan^{-1}\left(\frac{wb}{R_1 - \frac{t_f}{2}}\right) \\ \delta_2 = \tan^{-1}\left(\frac{wb}{R_1 + \frac{t_f}{2}}\right) \\ R = \sqrt{\left(R_1 + \frac{t}{2}\right)^2 + wb^2} \end{cases} \Rightarrow \begin{cases} \delta_1 = 29,94^\circ \\ \delta_2 = 23,63^\circ \\ R = 7009,4 \text{ mm} \end{cases}$$

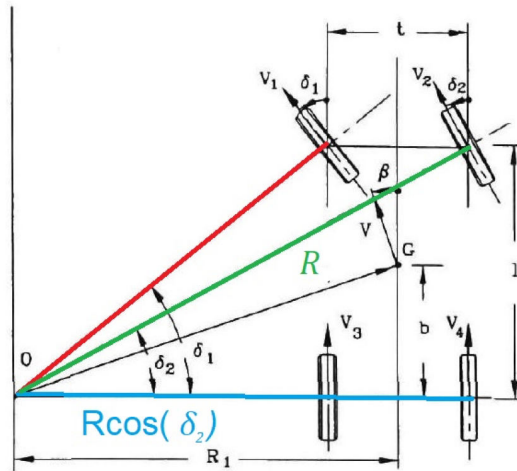


Figure 3: Kinematic steering

After that, knowing the sizing of kinematic steering angles, it's possible to use the

relevant data and parameters of the selected vehicle to evaluate the dimensions of front wheel arch.

Considering the parameter from table 1 (255/30 R19), it's possible to calculate the largest size of the front wheel arch:

$$Z = \frac{19}{2} \cdot 25,4 + \frac{30}{100} \cdot 255 = 317,8 \text{ mm} \quad (1)$$

2.2 Sizing of rear wheel arch

To evaluate the rear wheel arch, it's important to know that BMW 3 Series Sedan (2011) is a rear-wheel-drive vehicle.

Due to this assessment, the largest size of rear wheel arch can be calculate adding also the space for snowchains and dirt layer:

$$Z = \frac{19}{2} \cdot 25,4 + \frac{30}{100} \cdot 255 + 25 = 342,8 \text{ mm} \quad (2)$$

The contribution of snowchains and dirt layer have to add 25 mm to each side to compute the largest size of the wheel arch.

3 Side view analysis

In this section, the design of the side view of the vehicle is described. Firstly, the forces acting on the vehicle in different load conditions due to passengers and luggage are considered. Then, there is the calculation of the vertical stiffness of the front and rear axles in the full load condition and, finally, there is the measurement of the front and rear suspension deflections.

3.1 Load distribution

Firstly, it is fundamental to define the load conditions considered in this project. In A load condition, the vehicle is regarded as empty and loaded, so it is in its base conditions, ready to drive but without passengers or cargo. This weight includes all operating fluids (oil, coolant, brake fluids), full fuel tank and standard tools. In B load condition, the addition of the driver with a standard weight of 70 Kg and a luggage weight of 10 Kg is included. In C, it is taken into account also a passenger sitting beside the driver with the additional weight of the luggage. In D, E, and F, rear passengers with their weight and a luggage each.

The following dimensions are useful to calculate the force acting on front and rear axles due to passengers and luggage:

- df is the distance between the hip point of the driver and the front axle.

- dr denotes the distance between the hip point of the driver and the rear axle.
- pf represent the separation between the front axle and the hip points of rear passengers.
- pr measures the the distance between rear passengers and the rear axle

The empty mass of the vehicle is 1420 Kg and the weight distribution is 51:49. Consequently, the front and rear weight of the vehicle in empty condition is computed. Considering the B load condition, the weight of the driver acts on both axes in a different proportion. The reaction of the front and rear suspensions depends on the position of the front seat with reference to both axes. The reaction force of the front axle is:

$$\Delta F_f = \frac{dr}{wb} P \quad (3)$$

in which wb is the wheelbase and P is the passenger weight ($70kg \cdot 9.81N/Kg$) while the reaction force exerted by rear suspensions is:

$$\Delta F_r = \frac{df}{wb} P \quad (4)$$

In addition, we need to evaluate the reaction forces due to the luggage weight. The luggage distance is the same for each additional luggage of other passengers. Its contribution to the rear reaction force is:

$$\Delta F_{lr} = \frac{rl}{wb + rl} Pl \quad (5)$$

in which Pl is the luggage weight. While its contribution on the front axle is:

$$\Delta F_{lf} = -\frac{wb}{wb + rl} Pl \quad (6)$$

It is important to put in evidence the negative sign of this force. As a matter of fact, the axis on which luggage acts is located rearwards with respect to the rear axle. Consequently, it produces a reaction downwards on the rear axle and upwards on the front axle.

These values should be added to the reaction forces in the empty condition to evaluate the total reaction forces in the B condition. Considering the C load condition, a passenger aside the driver must be evaluated. On the side view, his position is located at the same distance from the front and the rear axes used for the driver's case. So, it is enough to double the weight acting on the front seats and on the luggage location because it is supposed that every passenger has a mass of 70 kg and each luggage of 10 kg.

In the D, E and F conditions, the vehicle includes also rear passengers. Rear passengers' row is located at a distance equal to pf from the front axle and to pr from the

rear axle. To compute forces acting on front axles, we use the formula:

$$\Delta F_f = \frac{pr}{w} P \quad (7)$$

and the force acting on the rear axle is:

$$\Delta F_r = \frac{pf}{w} P \quad (8)$$

In the table, there is the summary of the main distances used in the evaluations of load on front and rear axles.

It can be noticed that both hip points are located behind with reference to the center of gravity of the vehicle (df is at 1624 mm from the front axle, while the center of gravity is at 1400 mm). Consequently, when passengers' weight is applied, most of the efforts are related to the rear axle. In addition, this leads to a shift of the weight balance towards the rear side. This trend will increase in more heavy load conditions.

These measurements are presented in the side view of the vehicle in figure 9. Moreover, the rear passenger row is located very close to the rear axle, so the weight is transmitted mostly to it, while the reaction forces on the front suspensions will be low.

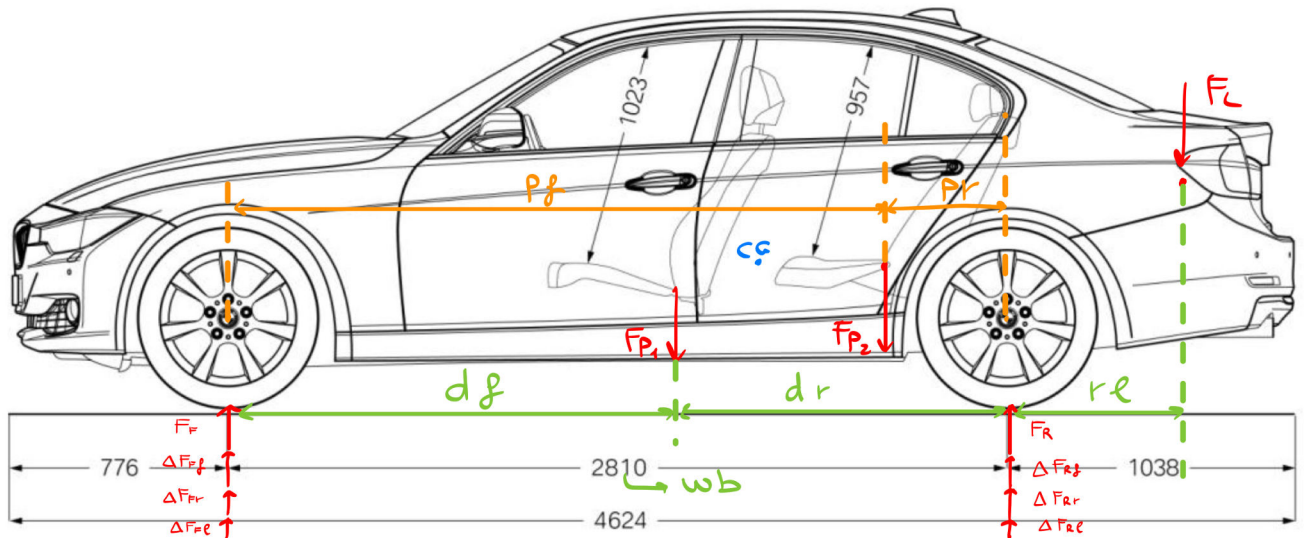


Figure 4: Caption

Table 2: Distances of passengers and luggage from front and rear axles in the BMW 3 series

Parameter	Value [mm]
df	1624
dr	1186
P	686.7
rl	519
Pl	98.1
pr	440
pf	2370

In table 3 it can be seen the number of front and rear passengers in different load standards. For each of them, reaction forces will be computed.

In table 4 there is a summary of the reaction forces on both axles under different load conditions. The columns represent forces on the front and rear axles due to passengers' weights (ΔF_f and ΔF_r) and due to luggage weights (ΔF_{fl} and ΔF_{rl}).

Table 5 provides an overview of the total weight acting on front and rear axles including the weight of the empty vehicle, luggage and passengers. It can be put into evidence that total rear weight increases more on the rear axle than on the front. According to these data, weight distribution can be easily evaluated. It is the ratio between total front weight and total rear weight at each load condition. As both passenger rows are located behind the center of gravity, the weight distribution shift to the rear by adding additional passengers. Consequently, it has a decreasing trend going from A to F load standard. In full load condition, the distribution is 45-55, so the weight is more shifted to the rear axle is more under stress.

Table 3: Load standards

Load condition	Front passengers	Rear passengers
A	0	0
B	1	0
C	2	0
D	2	1
E	2	2
F	2	3

Table 4: Load distribution under different conditions

Load condition	ΔF_f	ΔF_r	F_{rl}	F_{fl}
A	0	0	0	0
B	289.831	396.869	82.806	-15.294
C	579.663	793.737	165.612	-30.588
D	687.189	1372.911	248.418	-45.882
E	902.241	1952.085	331.224	-61.176
F	1224.819	2531.259	414.030	-76.470

Table 5: Total weight distribution under different conditions

Load condition	Total front weight	Total rear weight
A	7104.402	6825.798
B	7378.939	7305.473
C	7653.477	7785.147
D	7745.709	8447.127
E	7945.467	9109.107
F	8252.750	9771.087

3.2 Suspension stiffness in full load condition

In full load conditions, the suspension stiffness plays a crucial role in maintaining vehicle stability, ride comfort, and handling performance. Suspension stiffness, typically expressed as the spring rate k_s , determines how much the suspension compresses under a given load. When the vehicle is fully loaded with passengers and cargo, the vertical load on each wheel increases, causing greater suspension deflection. If the suspension is too soft, excessive sagging may occur, leading to reduced ground clearance, increased body roll, and compromised handling. On the other hand, if the suspension is too stiff, the ride becomes harsh, transmitting road irregularities directly to the cabin. Therefore, designing an optimal suspension stiffness for full load conditions requires balancing comfort and dynamic stability, ensuring that both front and rear suspensions support the vehicle weight efficiently while maintaining consistent tire contact with the road. The suspension stiffness k_s can be estimated using the natural frequency f of the vehicle, according to the relation $2\pi f = \sqrt{\frac{k_s}{m_s}}$. In which f is the natural frequency of the suspensions, k_s is the stiffness coefficient and m_{fl} is the mass of the vehicle in full load condition.

m_s is evaluated in F load standard, so in the most critical condition for the suspensions. f is the natural frequency which needs to be determined considering the fact that occupants of a vehicle have a different perception of comfort in relation to the amplitude and the frequencies of vibrations they are subject to. Below 1 Hz, the sensitivity of humans increases because motions of low frequency could excite the equilibrium organs inside the ear, leading to nausea and motion sickness. On the other hand, in the range 6-20 Hz, the level of comfort is high due to the resonance caused inside some internal organs.

For both these reasons, the vibration frequency must be maintained in the range between 1 and 6 Hz. Car manufacturers usually use a value of 1.5 Hz, which is adopted in the following calculations.

Given f , front and rear stiffness can be evaluated by using the full load mass, which is the mass of the vehicle in F load standard. In the BMW 3-series it corresponds to 1820 kg, which is the empty mass of 1420 kg with the addition of 5 passengers of 70 kg each and 5 luggage of 10 kg each. The mass of the vehicle acts differently on the

axles, so it must be multiplied by the weight distribution calculated before to find the value of front and rear sprung mass, which must also be divided by 2, because each axle has 2 wheels, so there are 2 springs on which forces act.

The following formulas represent the front and rear suspension stiffness:

$$k_f = 4\pi^2 f^2 m_{flfront} Wd/2 = 36,854.99 \text{ N/m} \quad (9)$$

$$k_r = 4\pi^2 f^2 m_{flrear} Wd/2 = 43,977.06 \text{ N/m} \quad (10)$$

in which wd is the weight distribution on axles, f is the natural frequency and m is the mass in F load standard.

3.3 Suspension deflection

The last part of this section deals with the evaluation of suspension deflection. Suspension deflection refers to the vertical movement of the wheel and axle relative to the body of a vehicle, which occurs during normal operating conditions and can approach maximum limits during impacts or body roll.

The main functions of a well-designed suspension system can be summarized as follows.

- Ride quality. A vehicle suspension has a function of providing an isolation by decreasing forces transmitted from the vehicle axle to the vehicle body. The accelerations of the sprung mass are used as a performance indicator.
- Suspension deflection limit. An excessive suspension bottoming should be avoided. Hence, the suspension stroke should be constrained to a prescribed level.
- Road holding. The wheels and the road should contact uninterruptedly. Moreover, the dynamical tire load should not exceed the static one.

For the design of a suspension system, it is crucial to compute its deflection due to the weight of passengers and cargo. In the previous paragraph, the elastic constant of front and rear suspensions has been measured. The case study was the F load standard, which is the full load condition, in which the vehicle is subjected to the maximum stress.

Starting from this point, the maximum deflections can be calculated by using Hook's law:

$$\Delta l_f = \frac{(F_{fl} + F_{fp})}{K_f} \quad (11)$$

$$\Delta l_r = \frac{(F_{rl} + F_{rp})}{K_r} \quad (12)$$

As table 6 shows, the rear axle is subjected to the higher load and so rear suspensions deflect more. The maximum deflection in the F load condition is 31.16 mm on the front axle and 66.9 mm on the rear axle.

This deflection is compatible with the distance between the rear wheel arch and rear tires. This is important to avoid interferences if the car is in full load conditions.

Table 6: Front and rear deflection under different load conditions

Load Condition	Front Deflection [mm]	Rear Deflection [mm]
A	0.000	0.000
B	7.449	10.907
C	14.898	21.815
D	17.401	36.868
E	22.821	51.920
F	31.159	66.973

4 Ground clearance

In this section, it will be explained how the ground clearance angles of this BMW were estimated under minimum and maximum load conditions. First, the previously calculated front and rear suspension deflections were used to compute the angles under various load conditions. Finally, the total vertical travel of the suspension was then estimated.

4.1 Ground clearance angles

Ground clearance (also called ride height) is the height between the lowest point of the vehicle's undercarriage (excluding the wheels) and the ground. Basically, it will be valuable to the handling, stability, and drive quality of the car. For example, the lower the ground clearance, the sporty but unsafe it is to use; on the other hand, the higher the ride height, the safer but adversely affect the aerodynamic features.

Firstly, after the suspension deflections computation in the previous chapter, it has calculated the variation of the ground clearance angles on each loading condition using the arc tangent function between the length of the wheelbase and deflections of the vehicle. The grades have computed as follows. Angles were determined by taking the difference between the rear and front suspension deflections, dividing it by the wheelbase, and applying the inverse tangent to the resulting ratio, as illustrated in Figure ???. As shown, it has started from 0 ° in load condition A (curb load) to 0.73 ° in load condition F (full load); mainly, the vehicle in curb condition presented a slight grade on the rear side with respect to the front side. Otherwise, in full load, it showed that the rearwards is more balanced with the forward. Furthermore, although it is not depicted in the figure, the center of gravity (CoG) in state A will be located slightly above, and ahead of, the CoG in state F.

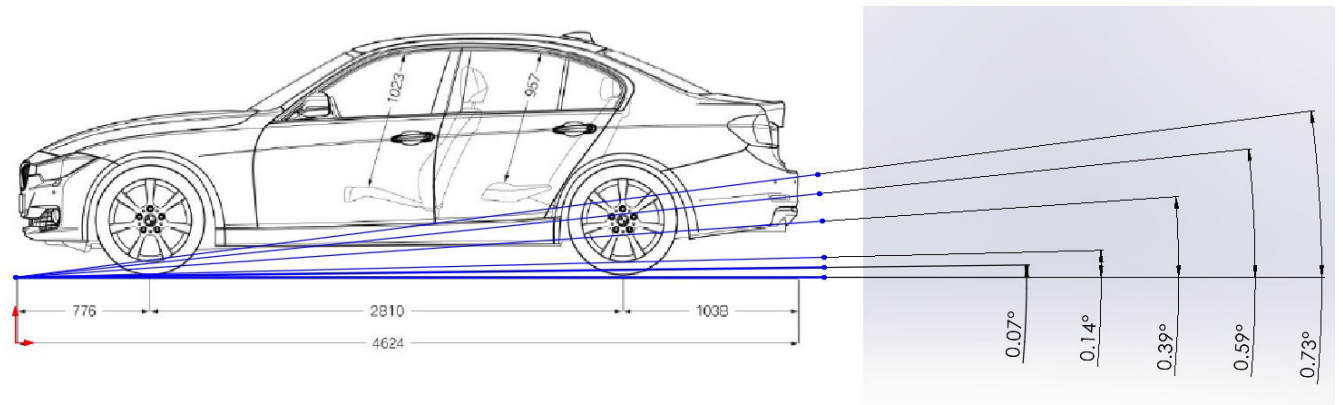


Figure 5: Ground clearance at different load conditions

5 Results and Discussion

5.1 Front wheel arch

Once the size of front wheel arch have been calculated, it is possible to quote the latter in top view (showing the maximum steering angles) and side view, considering the displacement of 30 mm the ground clearance (from the side view) and an additional travel of 67 mm for the suspension in top view.

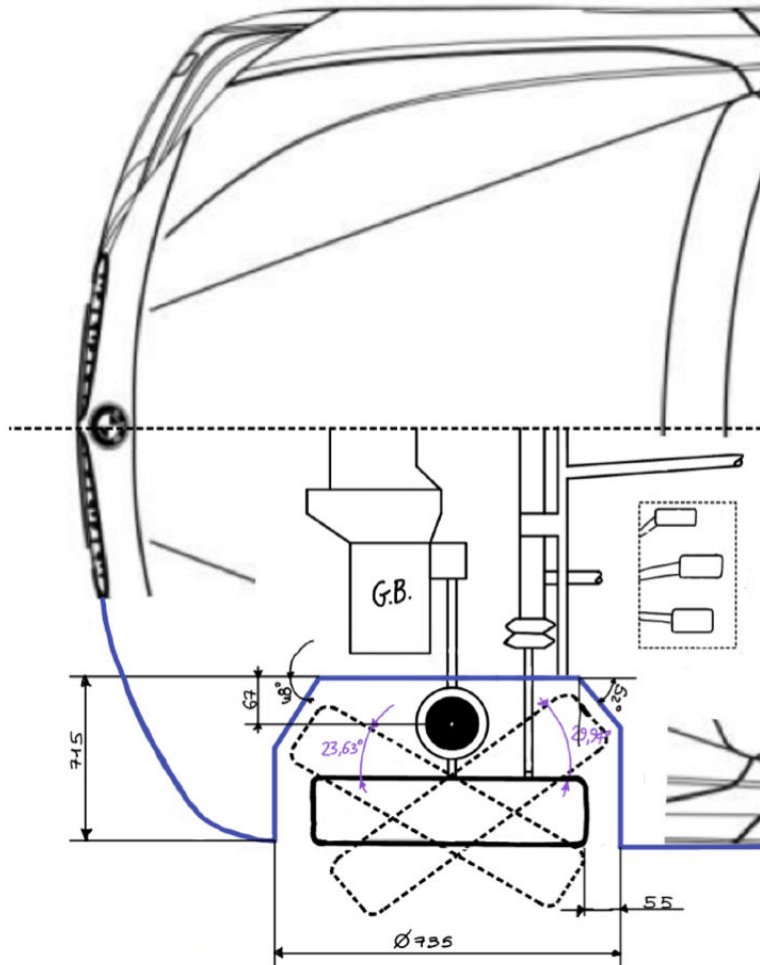


Figure 6: Top view of front wheel arch

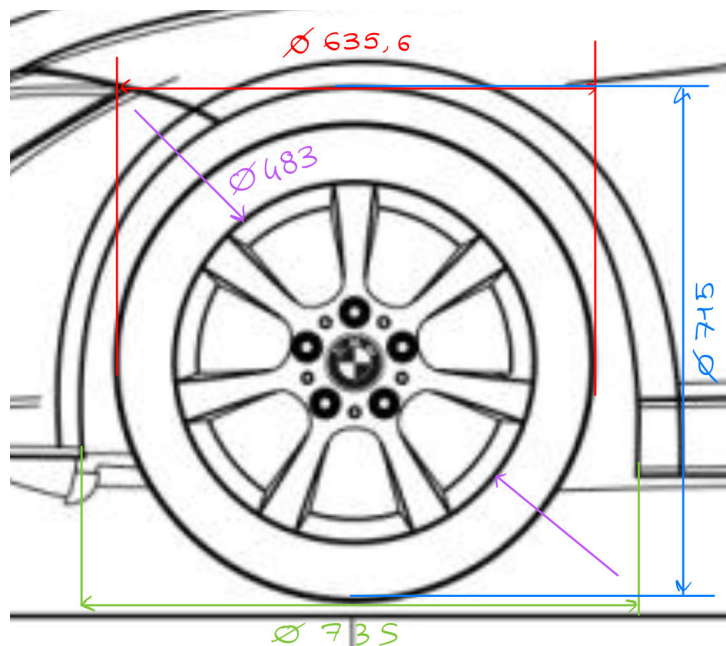


Figure 7: Side view of front wheel arch

5.2 Rear wheel arch

For the rear wheel, to quote the rear wheel arch it is necessary to consider, as well as the displacement of ground clearance and additional travel for suspension, the known gap of 25 *mm* for snowchains and dirt layer for each side. So an additional gap about 50 *mm*.

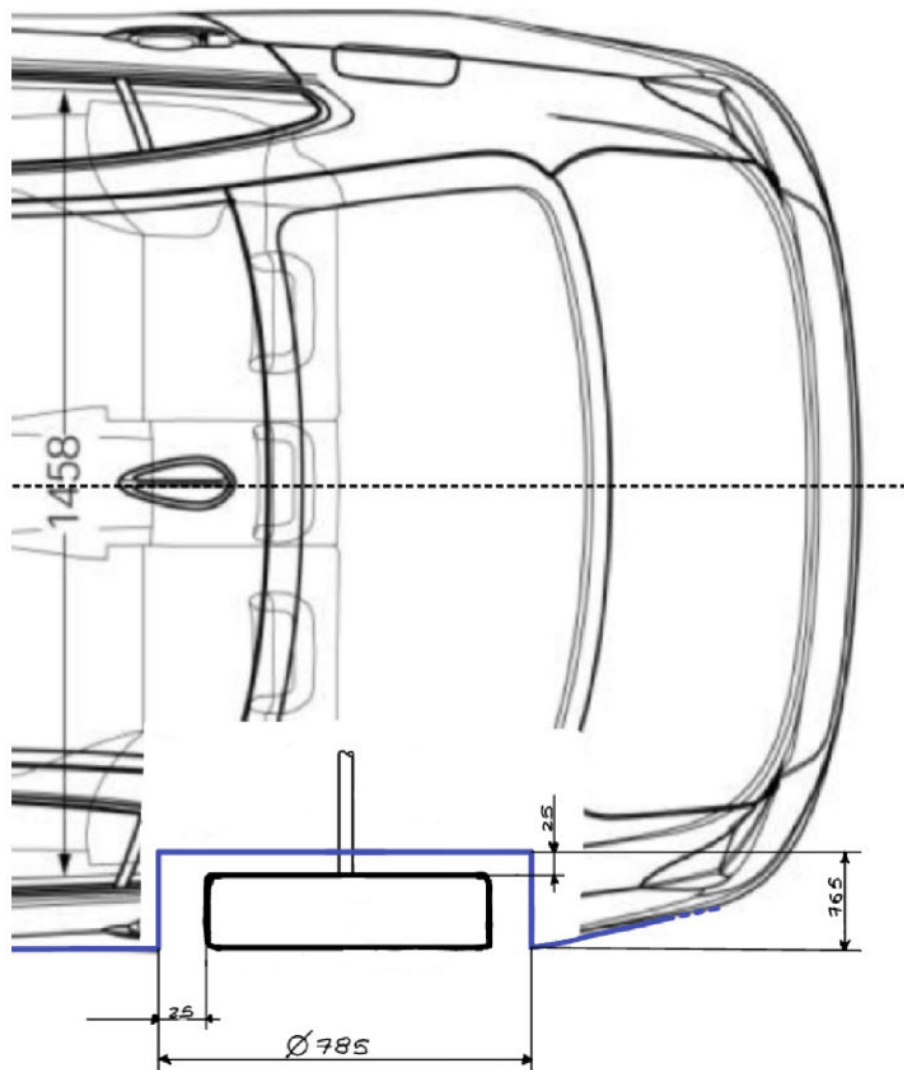


Figure 8: Top view of rear wheel arch

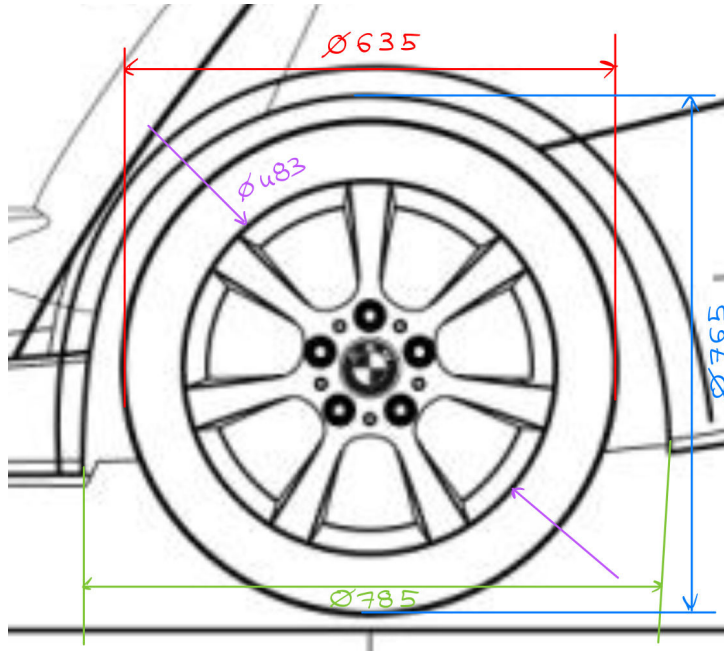


Figure 9: Side view of rear wheel arch

In *Figure 8* and *Figure 9*, it is possible to see the difference of quote from *Figure 6* and *Figure 7* due to the additional space necessary to snowchains and dirt layer: indeed it has been added 50 *mm* in total for the rear wheel arch.

5.3 Ground clearance

For this final step, an approximate height of the ground clearance of the vehicle was performed. Specifically, under fully loaded condition (state F), with rear and front suspension deflections of 31.159 mm and 66.973 mm respectively, a pitch angle of 0.73°, and an additional vertical suspension travel of approximately 30 *mm*, the ride height was estimated to be around 150 mm. This approximation provides a reasonable representation of vehicle clearance in real operating conditions, taking into account both static and dynamic effects on the suspension system.

6 Conclusion

The procedure adopted in this report allows the dimensions of the front, rear wheel arches and the ground clearance of BMW 3 Series Sedan (2011).

This approach was devoted to practical operating conditions, including the need to fit snowchains (for this vehicle inserted in the rear wheel), the consideration for dirt layer and load variations ranging from an empty vehicle to full one with all the passengers and luggage. In conclusion, the results obtained allow us to understand how the sizing of these elements are fundamental to improve the stability, performance and safety of the vehicle and every cases are important to study to obtain the best possible result.



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Project 2

Battery dimensioning for Electric vehicles

Car Body Design

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1 Introduction

The primary objective of this report is the comprehensive design of a battery pack tailored for commercial electric vehicles. The methodology commences with a preliminary sizing phase to establish nominal voltage and energy requirements, followed by the strategic configuration of cell and module layouts. The study culminates in a detailed evaluation of three critical domains: the effectiveness of the thermal management system, the architecture of power and signal interconnects, and a comparative assessment of gravimetric and volumetric energy densities at both the component and system levels

2 Battery pack

2.1 Nominal voltage and nominal energy

The battery studied is about BYD Seal Long Range with the following nominal values:

Parameter	Value	Unit
Nominal Battery energy	82500	Wh
Nominal voltage	550	V

Table 1: Nominal voltage and nominal energy

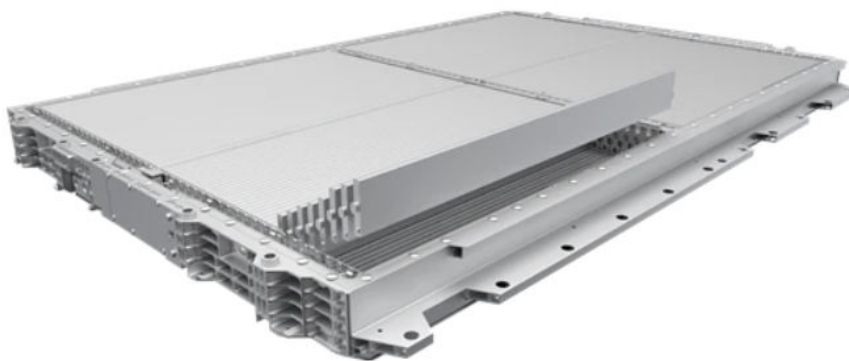


Figure 1: *BYD battery pack.*

Using this two parameters, it's possible to found the battery capacity of BYD Seal with following equation:

$$Battery\ capacity = \frac{battery\ energy}{nominal\ voltage} = \frac{82500}{550} = 150\ Ah \quad (1)$$

3 Battery layout

For the dimensioning of the BYD Seal Long Range battery pack, a Cell-to-Pack layout has been selected. Unlike traditional architectures that group cells into intermediate modules before assembly into the main enclosure, this design integrates the cells directly into the battery pack without structural functions. Furthermore, this layout doesn't have any subpacks. This choice is strategic and necessary due to the adoption of Lithium Iron Phosphate (LFP) chemistry. While this chemistry offers superior safety and cycle life, it inherently possesses a lower gravimetric energy density compared to other chemistries.

To compensate for this, the Cell-to-pack layout significantly increases volume utilization by removing passive components such as module casings, separate wiring harnesses, and support beams. This optimization increases the volumetric energy density by approximately 50% compared to conventional designs, allowing the vehicle to achieve the required Long Range capacity within the available physical footprint. Furthermore, in this configuration, the cells themselves provide structural integrity to the pack, creating a robust, honeycomb-like structure that enhances torsional stiffness while minimizing parasitic weight. In the following table, it's possible to see the main characteristics of BYD's Seal battery:

Parameter	Value	Unit
Mass battery m_b	580	kg
Volume battery V_b	250	L
Dimension battery pack	1308x1990x110	mm

Table 2: Battery dimension

3.1 Cell type

For this type of cell, a prismatic cell has been chosen. This geometry has high cost, low standardization and a difficult thermal management, but it is superior to cylindrical options for this application primarily due to its packaging efficiency; the rectangular shape allows the cells to be stacked with minimal void space, maximizing the volume efficiency. Furthermore, the prismatic design features a rigid outer casing (typically aluminum) which is essential for the BYD Seal's specific layout. Unlike soft pouch cells, which require external frames for support and compression, these rigid prismatic cells can serve as load-bearing structural beams. This capability allows the cells to contribute directly to the pack's mechanical integrity and torsional stiffness, eliminating the need for heavy internal bracing and fully enabling the weight-saving potential of

the Cell-to-pack design. This cell type of BYD Seal Long Range has the following properties:

Parameter	Value	Unit
Cell voltage	3.2	V
Cell capacity	202	Ah
Cell mass	3.9	kg
Cell Dimension (l×h×w)	905×118×13.5	mm

Table 3: Cell parameters

3.2 Cell layout

Subsequently, it is necessary to define the module layout X_s Y_p of the battery, where X_s indicates how many cells are in series, whereas Y_p define how many cells are in parallel in the battery:

$$X_s = \frac{\text{Nominal voltage}}{\text{Cell voltage}} = \frac{550 \text{ V}}{3.2 \text{ V}} = 172 \quad (2)$$

$$Y_p = \frac{\text{Battery capacity}}{\text{Cell capacity}} = \frac{150 \text{ Ah}}{202 \text{ Ah}} = 1 \quad (3)$$

This configuration allows to describe the connection of the modules. As a consequence, the cells connection is 172S1P: 172 cells in series and 1 in parallel.

Once cell layout is obtained, it is possible to define the Active Volume AV , that is the total volume occupied by the cells:

$$AV = \text{Volume of 1 cell} \cdot \text{total of cells} = 1.4 \text{ L} \cdot 172 = 240.8 \text{ L} \quad (4)$$

3.3 Thermal management system

The thermal management system is engineered to keep the battery pack within the accepted working range of 20–30°C, preventing thermal runaway risks and reducing capacity degradation over time. For the BYD Seal, the chosen architecture utilizes a highly efficient indirect liquid cooling and heating system integrated with a wide-temperature range heat pump. This setup employs large cooling plates placed directly above the cells to maximize heat transfer efficiency. The system is capable of actively warming the battery in cold climates (under 0°C) and cooling it during high loads, ensuring the cells remain in their optimal thermal window regardless of ambient conditions. This approach aligns with current industry standards for high-performance EVs,

preferring indirect liquid cooling over immersion techniques due to its proven maturity and structural integration capabilities.

4 3D design of the pack

This section discusses the 3D design of the battery pack considered in this study. The structure was primarily based on the configuration illustrated in Figure 1, and the 3D design of the battery pack was developed using SolidWorks. Once the individual components were generated, they were assembled to create the complete battery pack.

4.1 Mechanical structure and battery management systems

Initially, the fundamental component of the battery pack (the Blade cell) was implemented. As depicted in Figure 2, the Blade cell has approximately a length of 905 mm , a height of 118 mm , and a thickness of 13.5 mm . In SolidWorks, the material selected for this cell is *aluminum 3003 Alloy*.

On both sides of the cell, there are terminals. The primary terminals correspond to the positive and negative polarities, while the others serve as connections for the battery management sensors, thermal sensors, and mechanical fixtures. Then, all of these terminals are connected to the bus bars.

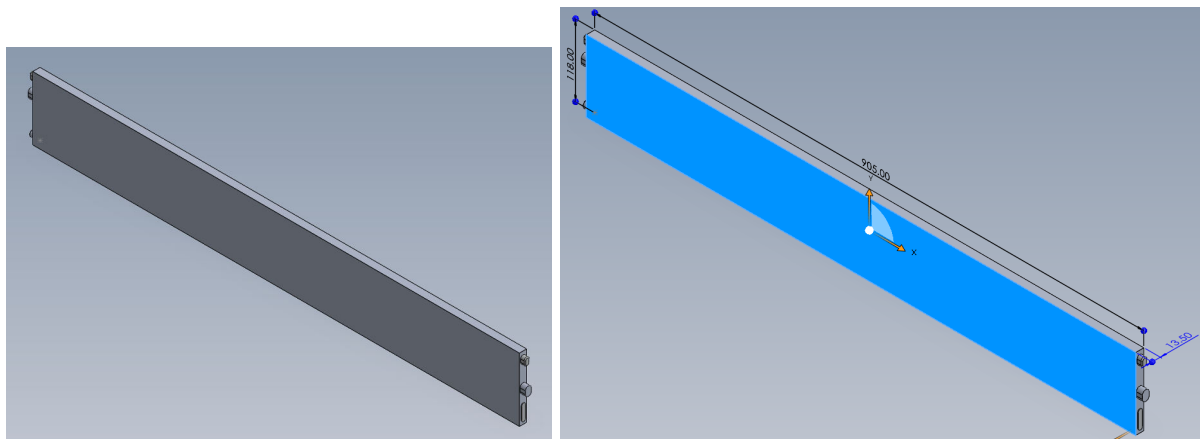


Figure 2: Blade cell and its dimensions.

The bus bars were built as shown in Figure 3. This component has a length of 2364.5 mm , a height of 118 mm , and a thickness of 2.5 mm . Mostly, the length and height represent the dimensions in which all the cells in series were placed. Moreover, the bus bar serves as the connection to the cells in series inside the battery case.

The holes in the middle of the bus bar are intended to accommodate the positive and negative terminals of the cells. This configuration allows for more compact series connections, improved thermal distribution, and more space-saving efficiency. The remaining holes serve different purposes: the upper ones are used for mechanical

fixtures, while the lower ones are used for connections to the battery management system (*BMS*) and thermal sensors. The material of this component was taken the same as the cells (**3003 Alloy**).



Figure 3: *Bus bars on left-side (upper), on right-side (middle), and their dimensions (lower).*

After that, the cooling system was created. As depicted in Figure 4, this component has a length of *2365 mm*, a height of *905 mm*, and a thickness of *3 mm* plus the serpentine thickness of *2 mm*.

In this case, it was tried to design a cold plate for liquid cooling. Generally, the coolant consists of water or glycol-based mixtures. The material selected for this cooling system is **Polyetheretherketone (PEEK)**.

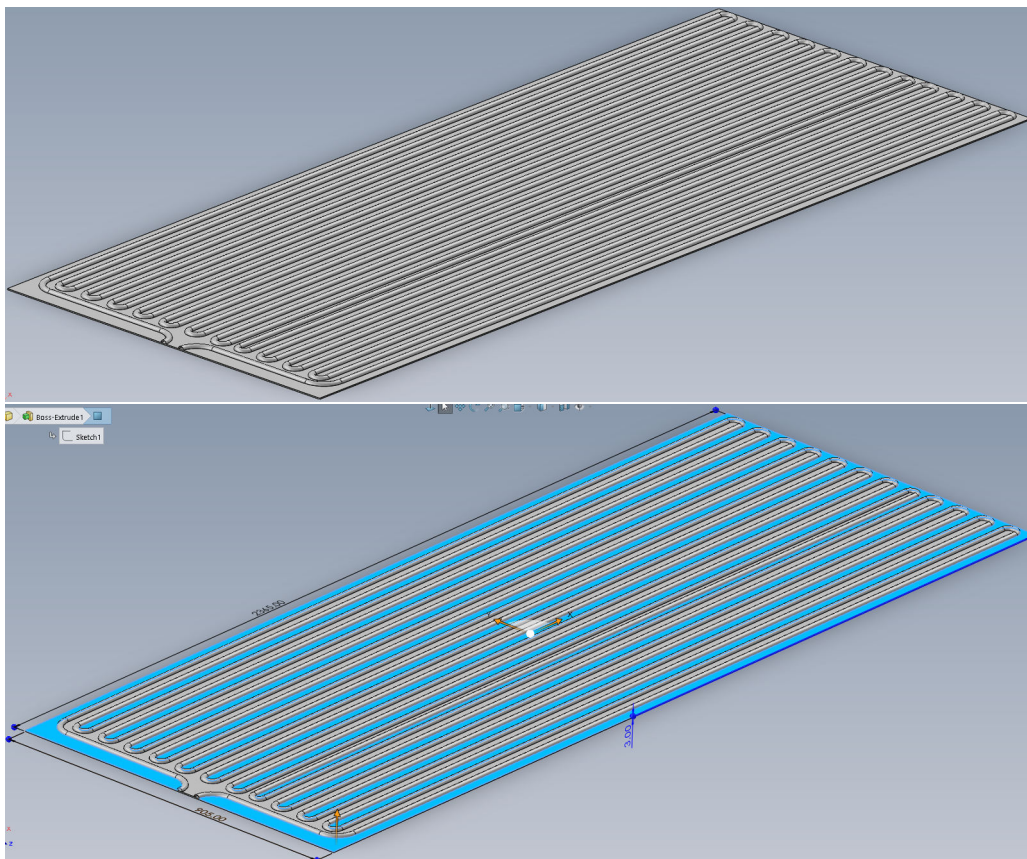


Figure 4: *System of cooling and its dimensions.*

Another component that it has created is the upper and lower cover of the battery pack. This component consists of a flat plate with a length of 2364.5 mm , a height of 925 mm , and a thickness of 3 mm . It is positioned above the cooling plate and below the cells and serves to protect both the cell module and the cooling system. For both upper and lower covers, the same cell material (**3003 Alloy**) was assigned.

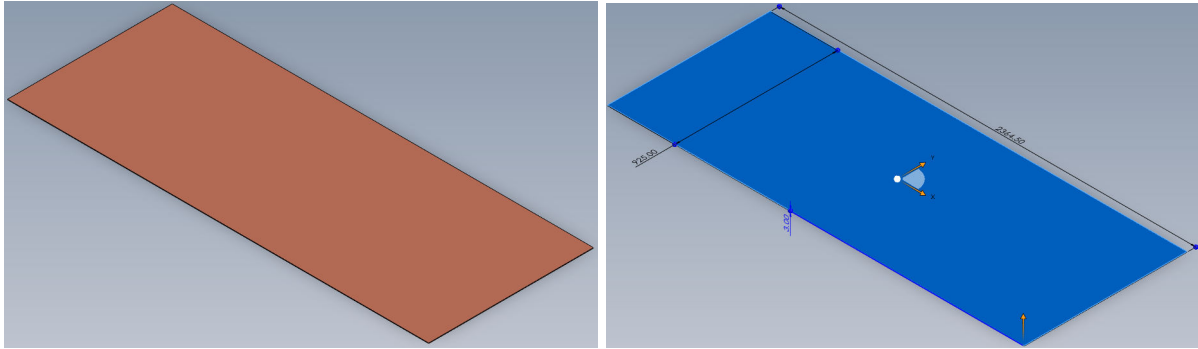


Figure 5: *Upper and lower cover and its dimensions.*

The last two fundamental components that were developed are the battery case and the top-cover of the case. As shown in Figure 6, the geometry of the battery case was designed to replicate the configuration illustrated in Figure 1. The internal dimensions were defined to accommodate the series-connected cells, the bus bars, and the lower cover. Similarly, the top cover was designed considering the space occupied by the upper cover and the cooling system below it. Furthermore, for mechanical fixtures, the hole dimensions were defined as *M14* for the lower part of the case and *M18* for the other holes, also taking into account the fixtures on the top-cover. The material that was considered for the two is aluminum **3003 Alloy**.

Although not visible in Figure 6, the electrical connectors are located on the bottom side of the battery case and are shown in Figure 7.

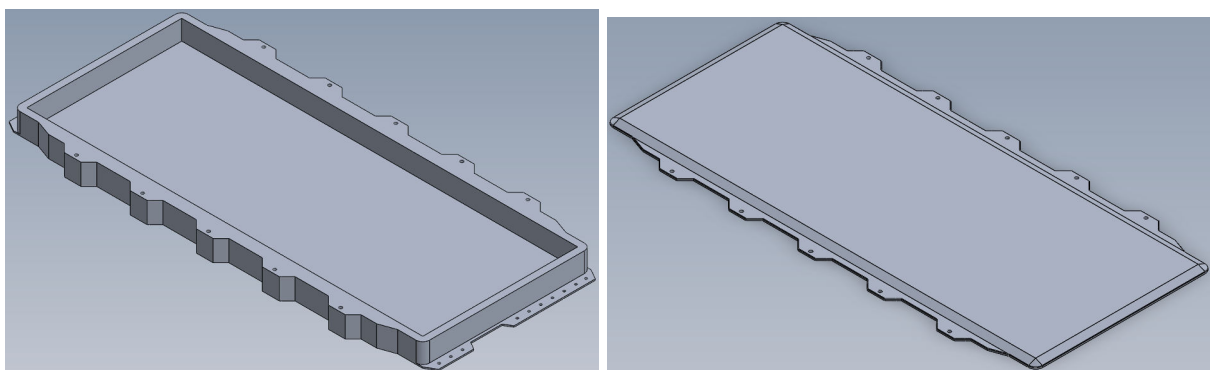


Figure 6: *Battery case (left) and the top-cover (right).*

4.2 3D completed design

Finally, once all the components were defined, the final result was obtained, as shown in the Figure below.

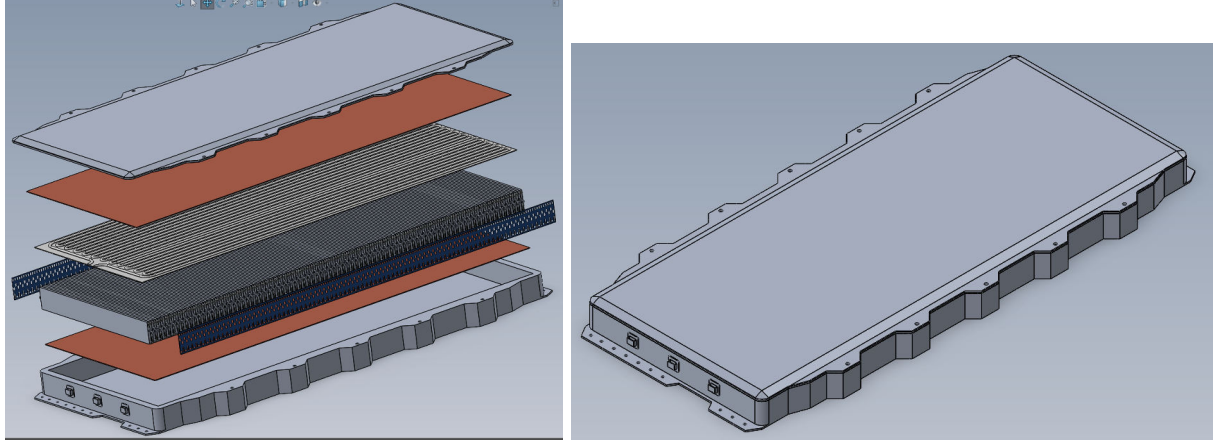


Figure 7: *Battery pack disassembled (left-side), and assembled (right-side).*

5 Resulting parameters

In this section, some important parameters of the battery pack will be evaluated to analyse the utilisation of the energy provided. It is fundamental to evaluate the amount of energy available per unit mass, which is referred to as *Gravimetric energy density*, or specific energy. It is typically expressed in Watt-hours per Kilogram (WH/kg) or Megajoules per kilogram (MJ/kg).

The gravimetric density expresses the capacity of the battery in terms of energy storage. Since batteries need to store energy for long periods of time, it is essential to use materials that have a great gravimetric energy storage capacity. However, this value is also used to define the energy content of the fuel. Gravimetric energy density is described by the following equation:

$$\rho_{EG} = \frac{E_b}{m_b} \quad (5)$$

where:

- E_b is the available energy expressed in Megajoules;
- m_b is the mass of the battery pack, expressed in kilograms.

If a material has a high energy density, it means that it is a light energy storage system, reducing the overall weight of the vehicle. In addition, there is the potential to increase performance in weight sensitive applications.

However, gravimetric energy density does not provide a complete description of the storage capacity of a battery because it does not take into account the volume occupied by it.

For this reason, it is important to introduce another parameter, called *volumetric energy density*.

Volumetric energy density is the total nominal energy of the battery divided by the volume it occupies. In other words, it defines the amount of energy stored per unit of volume. It is generally expressed in Watt-hours per Litre (Wh/L) or in Megajoules per litre (MJ/L).

It can be obtained by a simple calculation:

$$\rho_{Ev} = \frac{E_b}{V_b} \quad (6)$$

where:

- E_b is the total pack energy expressed in Wh;
- V_b is the battery pack volume expressed in Litres.

The last important parameter to evaluate is the *volume utilisation efficiency (VUE)*. It is the ratio between the volume occupied by the active material of the battery (battery cells) and the total volume of the battery pack, which also includes structural parts, such as the case, cooling, wiring and empty space. It can be described by the following equation:

$$VUE = \frac{AV}{V_{tot}} \times 100\% \quad (7)$$

in which:

- AV is the active volume in Litres;
- V_{tot} is the total volume of the battery pack.

A high value of the Volume utilisation efficiency means that a great part of the pack's volume is composed by energy-storing material. Consequently, the volumetric energy density will be high, as well.

It is also important to have a high value of *VUE* to reduce inactive space without necessarily changing cell chemistry. This fact is crucial in EVs where space is limited, so we need to maximise stored energy per volume.

5.1 Theoretical data

The purpose of this section is to calculate the parameters introduced previously, by using equations, and then to compare them with the mass and volumes resulting from the CAD simulation (in SolidWorks).

From the datasheets, the capacity of the battery pack is 82.5 *KWh*, which is also the energy that the battery can store. The mass of the battery pack is equal to 580 *kg* and the volume of the battery pack is 250 *L*.

To compute the volume occupied by the cells, their dimensions are needed. A single prismatic cell of the BYD Seal is 905 *mm* long, 118 *mm* wide and 13.5 *mm* deep. Consequently, the volume of a single cell is:

$$V_{cell} = (905 \cdot 118 \cdot 13.5) \text{ L} = 1.4 \text{ L} \quad (8)$$

Since the pack is composed of 172 cells, the total volume occupied by the cells is:

$$1.4 \text{ L} \cdot 172 = 240.8 \text{ L} \quad (9)$$

It is now possible to calculate the gravimetric energy density, the volumetric energy density and the volume utilisation efficiency. The gravimetric energy density is:

$$\rho_{E_G} = \frac{E_b}{m_b} = \frac{82\,500 \text{ Wh}}{580 \text{ kg}} = 142.24 \text{ Wh/kg} \quad (10)$$

The volumetric energy density is:

$$\rho_{E_v} = \frac{E_b}{V_b} = \frac{82\,500 \text{ Wh}}{250 \text{ L}} = 330 \text{ Wh/kg} \quad (11)$$

The volume utilisation efficiency is:

$$VUE = \frac{V_a}{V_{tot}} \times 100 \% = \frac{240.8}{250} \times 100 = 96.32 \% \quad (12)$$

5.2 CAD data

Now the parameters described in the previous subsection will be computed for the CAD model realised in Solidworks.

To evaluate the mass of the battery pack, the material of each component must be defined. The cells are made of Lithium-iron-phosphate LFP and their casing is made of aluminium alloy. For simplicity, the cell part is considered to be entirely composed of 3003 alloy.

The same material characterises the cases, the covers and the buses. Aluminium 3003 is one of the most widely used non-heat-treatable aluminium alloys. It is essentially pure aluminium (AA1100) with added manganese, which improves strength while keeping excellent formability and corrosion resistance. This material has also an excellent workability, a very good resistance to corrosion and a good weldability, so it is perfect for this application.

The main properties of 3003 aluminium alloy are summarised in the following table:

Property	Value	Unit
Elastic modulus	69 000	N/mm ²
Poisson coefficient	0.33	–
Density	2 700	kg/m ³

Table 4: Mechanical properties of the 3003 alloy

The binder is realised in polyetheretherketone (PEEK). PEEK is a semi-crystalline, high-performance engineering thermoplastic belonging to the PAEK polymer family. It is used when you need high strength, high temperature capability, chemical resistance, and long-term durability in aggressive environments.

The main properties are:

Property	Value	Unit
Elastic modulus	3 900	N/mm ²
Poisson coefficient	0.40	–
Density	1 310	kg/m ³

Table 5: Mechanical properties of the PEEK

Starting from these data, the mass property function of Solidworks can measure the mass and volume of the battery pack. Then, the energy density and the volume utilisation efficiency will be computed. The results are:

$$m_b = 665.64 \text{ kg}$$

$$V_b = 250.47 \text{ L}$$

The following calculations provide data, respectively, about gravimetric energy density, volumetric energy density and volume utilisation efficiency: The gravimetric energy density is:

$$\rho_{E_G} = \frac{E_b}{m_b} = \frac{82\,500 \text{ Wh}}{665.64 \text{ kg}} = 123.94 \text{ Wh/kg} \quad (13)$$

The volumetric energy density is:

$$\rho_{E_v} = \frac{E_b}{V_b} = \frac{82\,500 \text{ Wh}}{250.47 \text{ L}} = 329.38 \text{ Wh/kg} \quad (14)$$

The volume utilisation efficiency is:

$$VUE = \frac{V_a}{V_{tot}} \times 100 \% = \frac{240.8}{250.47} \times 100 = 96.13 \% \quad (15)$$

To compare these results, the relative errors are the following:

$$\Delta\rho_{E_g} = 18.3 \%$$

$$\Delta\rho_{E_V} = 0.6 \%$$

$$\Delta VUE = 0.2 \%$$

It is evident that the main source of error is the value of the mass, due to some simplifications relative to the material choice and to the lack of information about the correct alloy of each component.

6 Conclusion

This report successfully presented the complete design of a battery pack tailored for the BYD Seal Long Range. The project started with the preliminary sizing to define the main requirements, such as the 82500 Wh energy battery and the 550 V voltage. From there, the work moved to the detailed 3D design of the system.

One of the main decision in this project was the choice of the Cell-to-Pack layout combined with LFP chemistry. Since LFP batteries are characterized by high safety but lower energy density, a strategy to optimize space utilization was required. By eliminating intermediate modules and integrating the cells directly into the pack, this limitation was effectively overcome. This configuration maximizes the active volume, enabling the vehicle to achieve the required long driving range while ensuring the structure remains strong and robust.

The creation of the 3D model in SolidWorks served to verify these theoretical ideas. The design included all the essential components, such as the bars and the indirect liquid cooling system, which is necessary to keep the battery temperature between 20 °C and 30 °C. The simulation confirmed that the layout is feasible and fits perfectly within the vehicle's constraints.

Finally, the comparison between the calculations and the CAD model showed excellent results regarding space management. The volume utilisation efficiency was almost identical to the theoretical prediction. However, the weight in the simulation was higher than expected. As explained in the report, this difference is due to the material choice in the software: we used standard aluminum for simplicity, while the real manufacturer uses lighter, optimized alloys. Overall, the project demonstrates that the Cell-to-Pack technology is the key to making efficient and safe batteries for modern electric vehicles.



**Politecnico
di Torino**

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Project 3

**FEM modeling and modal analysis of a cantilever
beam using MATLAB**

Car body design and aerodynamics

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1 Introduction: description of the problem

The aim of this project is to analyze the fundamental mode of vibration of a clamped cantilever beam hit by a hammer using modal analysis and to understand the influence of the elastic modulus and damping coefficient on the frequency response of the structure, and to create a mechanical FEM model from experimental results obtained using an accelerometer.

1.1 Description of the system

The cantilever beam is clamped at one end and is subjected to an impulse produced by an hammer at a distance of 100 mm from the clamped end. Its main geometric and mechanical characteristics are the following:

- Beam length: $L = 280 \times 10^{-3} \text{ m}$
- Beam width: $b = 30 \times 10^{-3} \text{ m}$
- Beam thickness: $h = 3 \times 10^{-3} \text{ m}$
- Material: Aluminum AISI 2024
- Material density (aluminum): $\rho = 2700 \text{ kg/m}^3$
- Young's modulus (aluminum): $E = 64.5 \times 10^9 \text{ N/m}^2$
- Poisson's ratio: $\nu = 0.33$
- Shear modulus: $G = \frac{E}{2(1 + \nu)}$
- Number of finite elements: $n_{\text{el}} = 28$

1.2 Assumptions

Before starting with the FEM model, it is important to define the assumptions that have to be made regarding the material and how it resonates:

- The material is **homogeneous and isotropic**, so its properties (Young modulus, density, Poisson ratio) are the same in every direction;
- The **bending on XZ and YZ are decoupled**. Thus, the bending acting on one plane, doesn't affect the bending on the other;
- To simplify the analysis, the equivalent **2D system** is studied.

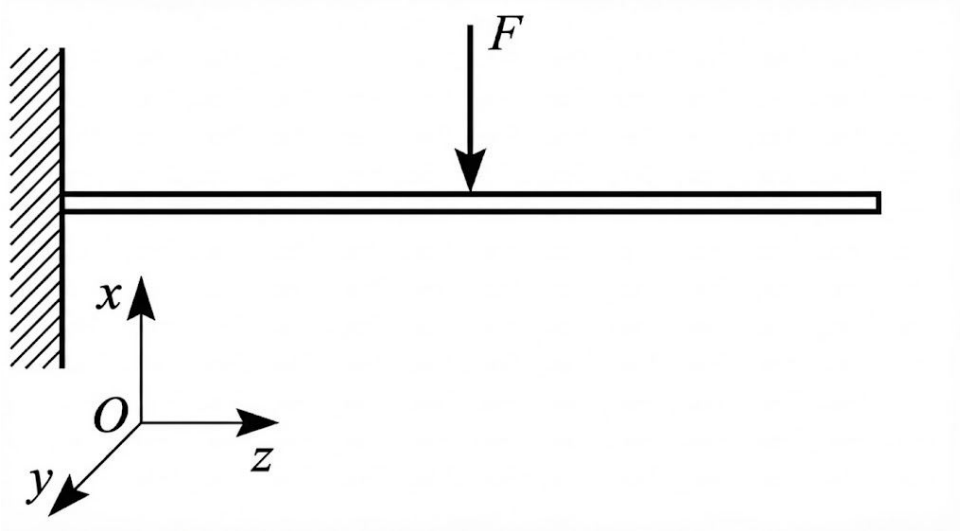


Figure 1: Representation of the cantilever beam during the experiment.

The last assumption is valid in cases where two dimensions are orders of magnitude larger than the third. This is particularly true in this beam, where the length is roughly one hundred times larger than the thickness, and the width is ten times larger than the thickness.

1.3 Numerical model

To define the FEM model the beam needs to be discretized using equally spaced nodes. In this case, the number of elements is 29, each of which has 2 degrees of freedom: the vertical translation along the x-axis u_x and the rotation about the y-axis ϕ_y .

Consequently, the total number of degrees of freedom is 58. Since the left end of the beam is clamped, the first two nodes have neither displacement nor rotation, thus, the system has only 56 degrees of freedom available.

2 Methodology

Each element of the beam has two nodes, with 2 degrees of freedom each: the vertical displacement and the rotation. The following equations set up the local dynamic model of the element:

$$\begin{Bmatrix} F_{x,i} \\ M_{y,i} \\ F_{x,i+1} \\ M_{y,i+1} \end{Bmatrix} = [\mathbf{M}] \begin{Bmatrix} \ddot{u}_{x,i} \\ \dot{\phi}_{y,i} \\ \ddot{u}_{x,i+1} \\ \dot{\phi}_{y,i+1} \end{Bmatrix} \quad (1)$$

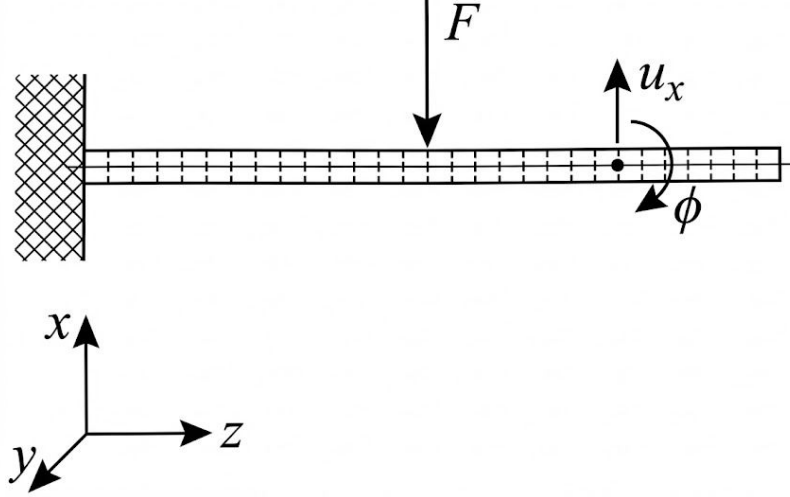


Figure 2: Discretization of the beam element and visual representation of DOF for each node.

$$\begin{Bmatrix} F_{x,i} \\ M_{y,i} \\ F_{x,i+1} \\ M_{y,i+1} \end{Bmatrix} = \frac{E I_y}{\ell(1 + \Phi)} [\mathbf{K}] \begin{Bmatrix} \dot{u}_{x,i} \\ \dot{\phi}_{y,i} \\ \dot{u}_{x,i+1} \\ \dot{\phi}_{y,i+1} \end{Bmatrix} \quad (2)$$

The equation 1 defines the relationship between nodal forces/moments and the generalized accelerations/velocities through a mass matrix M . This matrix produces inertial forces and expresses how the elements react at their nodes due to the application of an axial force F_x or a bending moment M_y .

The equation 2 expresses the same nodal forces/moments as a function of the nodal velocities using the stiffness matrix K . This refers to the elastic restoring forces that arise from bending stiffness.

The consistent mass matrix used for each 2-node beam element in a Timoshenko beam formulation is assembled from two contributions:

$$m_F = m_{F1_1} + m_{F1_2},$$

where

$$m_{F1_1} = \frac{\rho A \ell}{420(1 + \Phi_1)^2} \begin{bmatrix} m_1 & \ell m_2 & m_3 & -\ell m_4 \\ \ell m_2 & \ell^2 m_5 & \ell m_4 & -\ell^2 m_6 \\ m_3 & \ell m_4 & m_1 & -\ell m_2 \\ -\ell m_4 & -\ell^2 m_6 & -\ell m_2 & \ell^2 m_5 \end{bmatrix},$$

and

$$m_{F1_2} = \frac{\rho I_y}{30 \ell (1 + \Phi_1)^2} \begin{bmatrix} m_7 & \ell m_8 & -m_7 & \ell m_8 \\ \ell m_8 & \ell^2 m_9 & -\ell m_8 & -\ell^2 m_{10} \\ -m_7 & -\ell m_8 & m_7 & -\ell m_8 \\ \ell m_8 & -\ell^2 m_{10} & -\ell m_8 & \ell^2 m_9 \end{bmatrix},$$

The first term m_{F1_1} is associated with the linear momentum of the element and is related to the vertical displacement DOFs. This is due to mass distribution along the beam length, which resists transverse accelerations.

The second term m_{F1_2} is associated with the angular momentum of the section and it is related to the rotation DOFs. It represents the fact that the beam cross-section has a moment of inertia and, therefore, resists rotational accelerations.

In the Timoshenko formulation, it is important to include both terms to obtain a consistent dynamic behaviour and to consider rotation as an independent DOF.

Global mass M and stiffness K matrices are assembled by summing each element contribution using standard finite-element indexing. It is important to ensure that the mass and stiffness elements representing the displacement and rotation of the same node are summed together. The result is a 58×58 matrix which defines the dynamic behaviour of the system. For a cantilever beam, the first two DOFs are constrained due to the clamp, so the matrices used to solve the eigenproblem are:

$$M_c = M(3 : n_f, 3 : n_f)$$

$$K_c = K(3 : n_f, 3 : n_f)$$

The accelerometer mass is added as a concentrated mass at the DOF corresponding to the vertical displacement of the accelerometer node.

2.1 Modal analysis

The aim of the modal analysis is to determine the natural frequencies and mode shapes of the system. Mode shapes ϕ define how the structure deforms in each vibration mode, while natural frequencies ω represent the frequencies at which the beam vibrates.

To obtain these quantities, the eigenproblem must be solved using the following formula:

$$K_c \phi = M_c \phi \lambda \quad (3)$$

where:

- K_c is the global stiffness matrix
- M_c is the global mass matrix

- ϕ is the matrix of eigenvectors
- λ is the diagonal matrix of eigenvalues

From the eigenvalues, it is possible to obtain the angular frequencies of the system in [rad/s]:

$$\omega_i = \sqrt{\lambda_i}$$

They can also be expressed in Hz using this relationship:

$$f_i = \frac{\omega_i}{2\pi}$$

These are the dominant frequencies, that define the system's response behavior.

2.2 Damping model

This subsection describes how to obtain a damping matrix, that can be introduced in the finite element mode and used in dynamic simulations. Firstly, the mass and stiffness matrices must be expressed in the modal field by using the eigenvector matrix ϕ .

Eigenvectors are orthogonal with respect to M_c and K_c . so the resulting matrices M_m and K_m are diagonal. This coordinate transformation is useful as, in modal space, each mode behaves like an independent single degree of freedom system.

$$M_{modal} = diag(\phi^T M_c \phi)$$

$$K_{modal} = diag(\phi^T K_c \phi)$$

When the structural model is expressed in modal coordinates, each vibration mode behaves like an independent SDOF oscillator governed by:

$$M_{modal}\ddot{q}(t) + C_{modal}\dot{q}(t) + K_{modal}q(t) = Q_{modal}(t). \quad (4)$$

Because the matrices are diagonal, this equation decouples into N separate equations:

$$M_i\ddot{q}_i(t) + C_i\dot{q}_i(t) + K_iq_i(t) = Q_i(t), \quad (5)$$

where each index i corresponds to a single vibrational mode. Now, the critical damping can be evaluated. It is the minimum amount of damping that prevents a vibrational mode from oscillating. In modal analysis, the critical damping for each mode i is:

$$C_{c,i} = 2\sqrt{K_i M_i}$$

The actual damping of a structure is typically just a small fraction of the critical damping, so the model introduces a coefficient of tuning ζ that is used to rescale the damping

value. It is empirically evaluated, and it ensures light damping. ζ , in addition to the elastic modulus E , is the main parameter that can be used to tune the model.

$$C_{modal} = \zeta \cdot C_c$$

2.3 State space representation

The state space representation is a practical way to describe the dynamic behaviour of the system.

In order to avoid working with high order differential equations, the system is reformulated by defining:

$$X = \begin{bmatrix} q \\ \dot{q} \end{bmatrix}$$

Consequently, two first order equations can be defined:

$$\dot{x} = Ax + Bu \quad (6)$$

$$y = Cx + Du \quad (7)$$

in which the matrices A, B, C, D are defined as follows:

$$A = \begin{bmatrix} -M^{-1}C_{modal} & -M^{-1}K_c \\ I & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} M^{-1} \\ 0 \end{bmatrix}$$

$$C = A_{(2n_{hammer}-1, :)}$$

$$D = B_{(2n_{hammer}-1, :)}$$

In the previous equations:

- x is the **State vector** $[n]$, which defines the velocities and displacements of all DOFs;
- u is the **input vector** $[p]$, represented by the hammer excitation;
- y is the **output vector** $[q]$ and, in this case, is the response of the accelerometer;
- A is the **dynamic matrix** $[n \times n]$, which contains the dynamic of the system;
- B is the **input matrix** $[n \times p]$ and distributes the input into the states;

- C is the **Output matrix** $[q \times n]$ and selects the state variables subjected to measurement;
- D is the **Input/Output matrix** $[q \times p]$ and it is the direct feedback between input and output.

2.4 Bode plot

The best way to visualize the dynamic behavior of the beam is by using a bode plot. It is a frequency-response graph that shows how the system reacts to a sinusoidal input at different frequencies. It is composed of 2 plots: magnitude and phase. In this exercise, only the magnitude plot will be taken into consideration.

The **magnitude plot** shows the amplitude of the output relative to the input according to this relationship:

$$Magnitude [dB] = 20 \log_{10} |H(j\omega)|$$

The bode plot will be fundamental in the following section to compare experimental data with that from the FEM model.

3 Results and discussions

In this plot, the first six mode shapes of the nodes are presented relative to vertical displacement. To display only the vertical displacements, only the modes with an odd index will be shown. Each curve represents a single vibration mode from the fundamental (I) to higher degrees. As expected, the low modes are smoother and have fewer zero crossings, while higher order harmonics have more internal nodes, reflecting the increasing dynamic complexity.

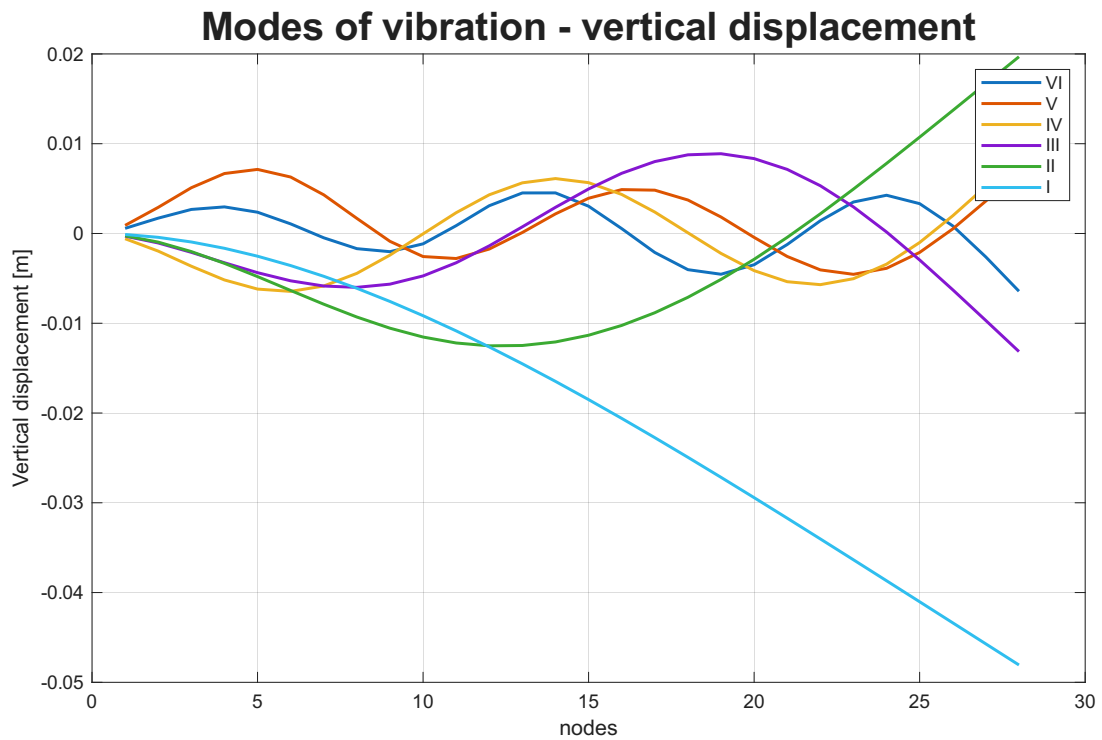


Figure 3: Modes of vibration of the vertical displacement degrees of freedom

In the following plot, the first six modes of rotation are displayed. In this case, modes with an even index are taken into account. Overall, the figure represents how the structure evolves when subjected to excitation, providing a clear overview of the dynamic evolution of the system.

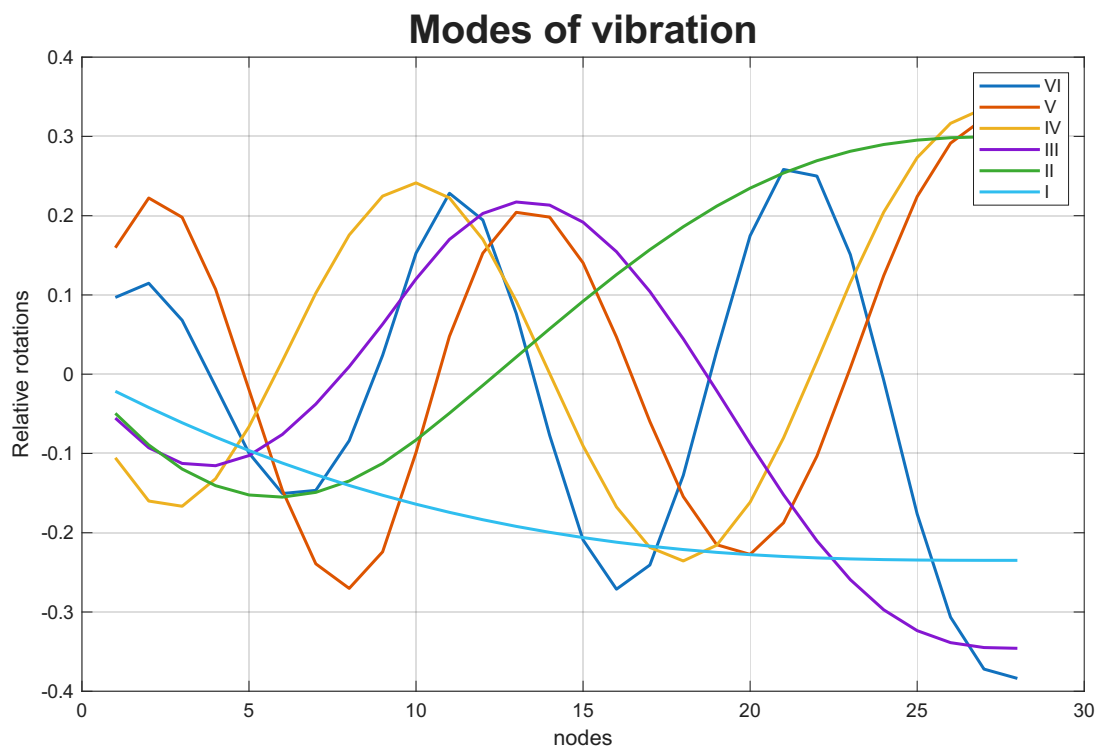


Figure 4: Modes of vibration of the rotational degrees of freedom

Finally, to evaluate the accuracy of the FEM model, it must be compared with experimental data. the bode plot below shows the magnitude-frequency spectrum of both sets of data. To adjust the model, there is the possibility to set two values: the damping ratio ζ and the elastic modulus E .

E affects the global stiffness matrix. Increasing the Young modulus shifts the curve upwards, while decreasing it shifts the curve downwards. the value E must be calibrated until the frequency peak of the model matches the experimental data.

The damping ratio ζ is mainly used to match the bandwidth and amplitude of the curve. If it increases, the line becomes lower and broader, while if it decreases, it becomes sharper.

The final values chosen after tuning are:

$$\zeta = 19 \cdot 10^{-3}$$

$$E = 64.5 \cdot 10^9 \text{ N/kg}$$

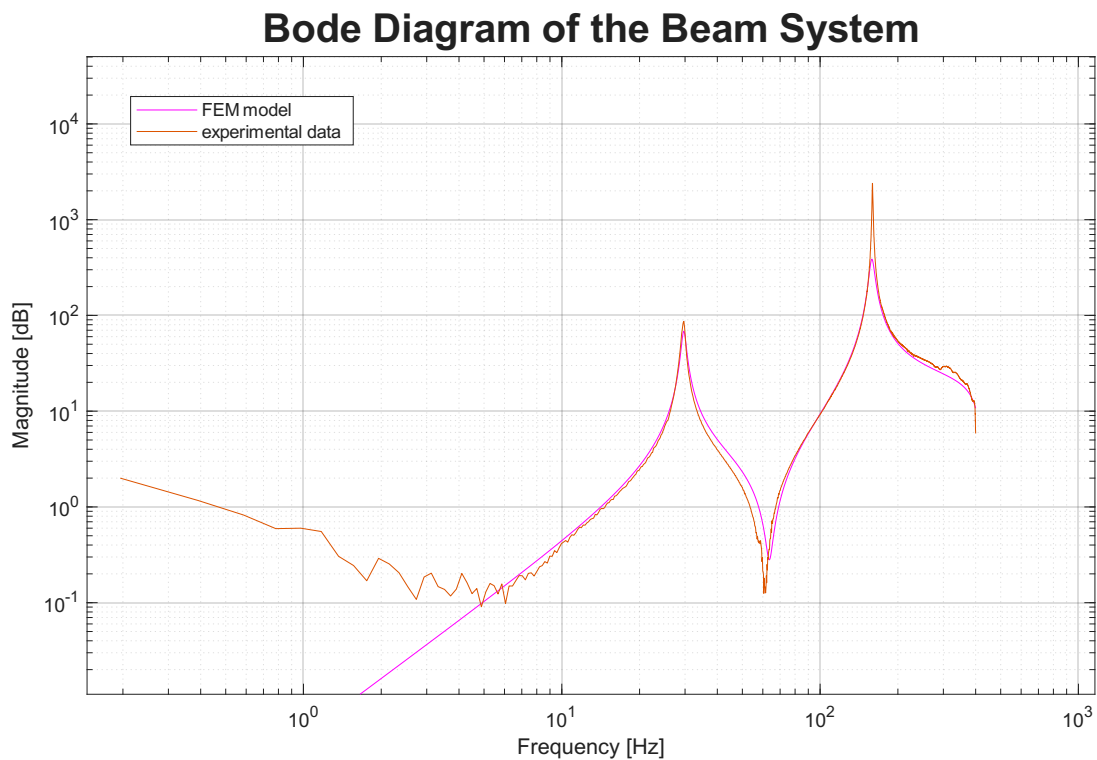


Figure 5: Bode plot representing the comparison between the experimental data and the FEM model frequency spectrum.

4 Conclusions

The model successfully described a calibrated dynamic of the system. The adjustment of E enables the alignment of the frequency peaks of the experimental data with those of the model, ensuring that it reflects the actual stiffness of the structure. The consequent tuning of ζ allowed for a correct match between the bode characteristics.

The model demonstrated consistent performance across eigenanalysis and state-space representation, confirming its robustness for predictive analysis.

5 MATLAB script

```
1 %%
2 % Car body design and aerodynamics
3 % a.a. 2025-2026
4 % Exercise 3 - FEM modeling and modal analysis of a cantilever beam
5 %
6
7 %%
8 format short e
9 clear
10 clc
11 close all
12
13 %% INIT
14 L          = 280e-3;                                %
15     ↪ [m]      Length of the beam
16 b          = 30e-3;                                %
17     ↪ [m]      Width of the beam
18 h          = 3e-3;                                  %
19     ↪ [m]      Thickness of the beam
20 ro         = 2700;                                  %
21     ↪ [kg/m^3] Density of the aluminum
22 massa_trave = ro*b*h*L;                             %
23     ↪ [kg]      Mass of the beam
24 E          = 64500e6;                                %
25     ↪ [N/m^2]   Young modulus of the aluminum
26 Poiss      = 0.33;                                  %
27     ↪ [-]       Poisson coefficient
28 G          = E/(2*(1+Poiss));                       %
29     ↪ [N/m^2]   Shear coefficient
30 n_el       = 28;                                    %
31     ↪ [-]       Number of elements
```

```

23 l = L/n_el; %
    ↪ [m] Length of a single element
24
25 m_acc = 18e-3; %
    ↪ [kg] Mass of the accelerometer
26 L_acc = 100e-3; %
    ↪ [m] Distance from the clamping point
27
28 L_hammer = 100e-3; %
    ↪ [-] Distance hammering point from the clamping end
29
30 n_n = n_el+1; %
    ↪ [-] Number of nodes
31 n_hammer = L_hammer/l+1; %
    ↪ [-] Node at which the hammer is applied
32 n_acc = L_acc/l+1; %
    ↪ [-] Node where the accelerometer is applied
33
34 A = b*h; %
    ↪ [m^2] Beam section area
35 I_y = b*h^3/12; %
    ↪ [m^4] Moment of inertia of the section area
36
37 chi = (12+11*Poisson)/10/(1+Poisson); %
    ↪ [-] Shear factor of a rectangular section
38
39 freq = 0.01:0.01:400; % [Hz] Frequency range of interest
40 %% Mass and stiffness matrix creation
41 % Behaviour in the plane orthogonal to plane of the beam node and reference node
42 PHI1 = 12*E*I_y*chi/G/A/l^2; %
    ↪ Coefficient which takes into account the shear deformation of the
    ↪ section in the plane orthogonal to plane of the beam node and reference
    ↪ node
43 m1 = 156+294*PHI1+140*PHI1^2;
44 m2 = 22+38.5*PHI1+17.5*PHI1^2;
45 m3 = 54+126*PHI1+70*PHI1^2;
46 m4 = 13+31.5*PHI1+17.5*PHI1^2;
47 m5 = 4+7*PHI1+3.5*PHI1^2;
48 m6 = 3+7*PHI1+3.5*PHI1^2;
49 m7 = 36;
50 m8 = 3-15*PHI1;
51 m9 = 4+5*PHI1+10*PHI1^2;
52 m10 = 1+5*PHI1-5*PHI1^2;

```

```

53
54 m_F1_1          = ro*A*l/(420*(1+PHI1)^2)*[ m1      1*m2      m3
    ↪ -1*m4
55              1*m2      1^2*m5      1*m4
    ↪ -1^2*m6
56              m3      1*m4      m1
    ↪ -1*m2
57              -1*m4      -1^2*m6      -1*m2
    ↪ 1^2*m5];
58
59 m_F1_2          = ro*I_y/(30*1*(1+PHI1)^2)*[m7      1*m8      -m7      1*m8
60              1*m8      1^2*m9      -1*m8
    ↪ -1^2*m10
61              -m7      -1*m8      m7
    ↪ -1*m8
62              1*m8      -1^2*m10      -1*m8
    ↪ 1^2*m9];
63
64 m_F              = m_F1_1+m_F1_2;
65
66 k_F              = E*I_y/(1^3*(1+PHI1))*[12      6*1      -12      6*1
67              6*1      (4+PHI1)*1^2      -6*1
    ↪ (2-PHI1)*1^2
68              -12      -6*1      12      -6*1
69              6*1      (2-PHI1)*1^2      -6*1
    ↪ (4+PHI1)*1^2];
70
71 M                = zeros(2*n_n);
72 MM               = zeros(2*n_n);
73 K                = zeros(2*n_n);
74 KK               = zeros(2*n_n);
75
76 for t=1:1:n_el
77 MM = zeros(2*n_n);
78 KK = zeros(2*n_n);
79 qq = 2*(t-1)+1;      % counter
80 qq1 = qq+3;          % counter
81
82 MM(qq:qq1,qq:qq1) = m_F;
83 KK(qq:qq1,qq:qq1) = k_F;
84 M=M+MM;
85 K=K+KK;
86 end

```

```

87
88 % Addition of the accelerometer mass at the proper node
89  $M(2*n\_acc-1, 2*n\_acc-1) = M(2*n\_acc-1, 2*n\_acc-1) + m\_acc;$ 
90
91 % constraints of the degrees of freedom
92  $nf = size(K, 1);$ 
93  $Mc = M(3:nf, 3:nf);$  %
94   ↪ Constrained mass matrix
95  $Kc = K(3:nf, 3:nf);$  %
96   ↪ Constrained stiffness matrix
97 %
98 %% MODE SHAPES
99  $[phi, lambda] = eig(inv(Mc)*Kc);$  % compute eigenvalues (lambda) and
100   ↪ eigenvectors (phi) solve  $Kc * Phi = Mc * Phi * lambda$ 
101  $eigenvalues = diag(lambda);$  % eigenvalues =  $\omega^2$ 
102  $\omega = sqrt(eigenvalues);$ 
103  $f\_natural = \omega / (2*pi);$ 
104
105  $N\_modes = 5;$  % number of modes to plot
106
107  $x\_nodes = 1 : 1 : 28;$  % positions along the beam
108  $figure(1)$ 
109
110 % Mode shape vertical displacement only (DOF odd indices)
111  $y\_mode = [ phi(1:2:end, i)];$ 
112
113  $plot(x\_nodes, y\_mode, '-','LineWidth',1.5);$ 
114  $hold on$ 
115  $xlabel('nodes');$ 
116  $ylabel('Vertical displacement [m]');$ 
117  $title('Modes of vibration','FontSize',20)$ 
118  $grid on; box on$ 
119  $end$ 
120  $legend('VI','V','IV','III','II','I')$ 
121
122  $figure$ 
123  $for i = 51:56$ 
124
125   % Mode shape vertical rotation only (DOF even indices)
126    $y\_mode = [ phi(2:2:end, i)];$ 

```

```

127     plot(x_nodes, y_mode, '-','LineWidth',1.5);
128     hold on
129     xlabel('nodes');
130     ylabel('Relative rotations');
131     title('Modes of vibration','FontSize',20)
132     grid on; box on
133 end
134 legend('VI','V','IV','III','II','I')
135 %% MODAL ANALYSIS
136
137 M_modal = diag(phi' * Mc * phi);           % transform M and k in modal field
138 K_modal = diag(phi' * Kc * phi);
139
140 Cc = 2* sqrt(K_modal .* M_modal);
141 zeta = 19e-3;                               % for tuning if you increase zeta
    ↪ peaks reduce
142 C_modal = diag(zeta * Cc);                  % compute C in modal coordinates
143
144 C_modal = phi' * C_modal * phi;             % convert C in original coordinates
145
146 %% State Space
147 M_inv= inv(Mc);
148 K_inv= Kc;
149 A          = [-M_inv*C_modal   -M_inv*Kc; eye(length(Kc))
    ↪ zeros(length(Kc))]; % dynamic matrix
150 B          = [M_inv; zeros(length(Kc))];
    ↪ % input matrix
    ↪ % accelerometer position
151 C          = A((n_hammer)*2-1,:);
    ↪ % output matrix
152 D          = B((n_hammer)*2-1,:);
    ↪ % direct input/output link matrix
153
154
155 system = ss (A, B, C, D) ; % state space
156 sys = system((n_hammer)*2-1);
157 w0 = freq * 2 * pi;
158 [mag, phase,wout] = bode (sys, w0); % plot at a single freq
159 mag1 = squeeze(mag);
160 figure
161 loglog(freq, mag1,'m'); % Plot in log-log scale
162 grid on;
163 title('Bode Diagram of the Beam System');

```

```

164
165 %% Comparison with Experimental data
166 % use the file Sq11_sn.mat to load experimental results
167 load('Sq11_sn.mat'); % Load experimental results for comparison
168 x= (FRF_Point2_Point1.x_values.start_value :
      ↪ FRF_Point2_Point1.x_values.increment :
      ↪ FRF_Point2_Point1.x_values.increment *
      ↪ (FRF_Point2_Point1.x_values.number_of_values-1));
169 y = abs(FRF_Point2_Point1.y_values.values);
170 hold on
171 grid on
172 loglog(x,y)
173 title ('bode plot')
174 xlabel ('Frequency [Hz]')
175 ylabel ('Magnitude [dB]')

```

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